



SS2 PHYSICS LESSON NOTES (THIRD TERM) WEEK 3

1. DEFINITION OF ELECTROMAGNETIC FIELD (EMF)

An electromagnetic field (EMF) is a physical field produced by electrically charged objects. It affects the behavior of charged particles in its vicinity.

The field consists of two components:

Component

Description

Electric field (E-field)

Produced by stationary charges. It exerts force on other electric charges in the field.

Magnetic field (B-field)

Produced by moving charges (current). It exerts force on other moving charges.

The electromagnetic field is governed by **Maxwell's equations**, which describe how electric and magnetic fields are generated and altered by each other and by charges and currents.

2. INTERACTION BETWEEN ELECTRIC AND MAGNETIC FIELDS

Electric and magnetic fields are inherently linked:

When an electric charge moves (i.e., a current flows), it produces a **magnetic field**.



A changing magnetic field induces an **electric field** (as described by Faraday's Law of Electromagnetic Induction).

This interaction forms the basis of **electromagnetic waves** (e.g., radio waves, light). In an electromagnetic wave, the electric and magnetic fields oscillate perpendicular to each other and to the direction of wave propagation.

3. LAWS OF ELECTROMAGNETIC INDUCTION

Law	Statement
Faraday's Law	Whenever there is a change in the magnetic lines of force, an EMF is induced. The strength of the induced EMF is proportional to the rate of change of the flux linked with the circuit.
Lenz's Law	The induced EMF is in such a direction as to oppose the motion or change producing it.

4. COMMON APPLICATIONS OF ELECTROMAGNETIC FIELDS IN EVERYDAY LIFE

Electromagnetic fields are pervasive in modern society. Some common applications include:

Electric Power Transmission: Transformers and high-voltage lines rely on alternating EMFs to transmit electricity.

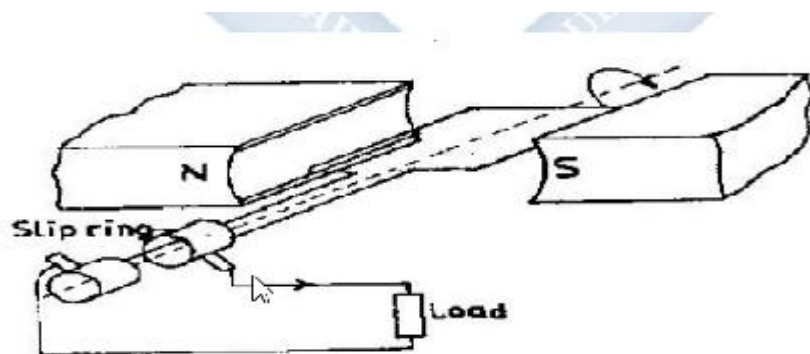
Household Appliances: Devices like microwaves, toasters, and induction cookers use EMFs for heating or operation.

Communication Systems: Radio, television, mobile phones, and Wi-Fi depend on electromagnetic wave propagation.

Medical Equipment: MRI scanners and diathermy machines use controlled EMFs for diagnostic and therapeutic purposes.

Security Systems: Metal detectors, RFID tags, and airport scanners operate using electromagnetic principles.

5. WORKING PRINCIPLES OF ELECTROMAGNETIC DEVICES



6.

Eddy Currents

Eddy currents are loops of electrical current induced within conductors by a changing magnetic field in the conductor. Eddy current flows in closed loops within conductors in planes perpendicular to the magnetic field. Eddy currents flow in such a direction as to oppose the motion of the conductor in the field.

Mutual Inductance



This is the flow of induced current or voltage in a coil due to an alternating current in a neighbouring coil.

5.1 Electric Motors

Convert electrical energy into mechanical energy. When current flows through a coil in a magnetic field, a force (Lorentz force) acts on the coil, causing it to rotate (based on **Fleming's Left-Hand Rule**).

5.2 Electric Generators

Convert mechanical energy into electrical energy using electromagnetic induction. Movement of a conductor (coil) in a magnetic field induces current (based on **Fleming's Right-Hand Rule**).

5.3 Transformers

Transfer electrical energy between circuits via mutual induction. They consist of primary and secondary coils wound around a magnetic core. A changing current in the primary induces a voltage in the secondary.

5.4 Loudspeakers

Convert electrical signals into sound. An alternating current in a coil causes vibration in a diaphragm attached to the coil, producing sound waves. This relies on the interaction between the coil and a permanent magnet.

5.5 Electromagnetic Relays



Electrically operated switches. A small current activates an electromagnet, which moves a switch to allow a larger current to flow through another circuit.

6. TECHNOLOGICAL APPLICATIONS OF ELECTROMAGNETIC FIELDS

6.1 Magnetic Resonance Imaging (MRI)

A diagnostic tool in medicine that uses strong magnetic fields and radio waves to produce detailed images of internal body structures. The magnetic field aligns the hydrogen nuclei in the body, and the radio waves disturb this alignment. The return to equilibrium emits signals used to construct images.

6.2 Induction Heating

Uses electromagnetic induction to heat electrically conductive materials (e.g., metals). An alternating current produces eddy currents in the material, generating heat due to electrical resistance. It is widely used in metallurgy and cookware (e.g., induction stoves).

6.3 Wireless Power Transfer

Transfers electrical energy without wires using oscillating magnetic fields (resonant inductive coupling). Common in wireless charging of phones, electric toothbrushes, and even electric vehicles.



7. SIMPLE DEVICES USING ELECTROMAGNETIC FIELDS

Device	Working Principle
Electromagnetic Relay	A switch controlled by an electromagnet. When current flows, the magnet pulls a contact to close/open a circuit.
Electric Bell	Works when an EM field pulls an armature, breaks contact, and allows repeated ringing action.
Loudspeaker	Coil in magnetic field vibrates diaphragm to produce sound in response to EM field changes.

8. ADVANTAGES OF USING ELECTROMAGNETIC FIELDS

- Enable contactless control of device
- Allow wireless communication
- Efficient energy conversion (motors/generators)
- Used in high-speed, non-contact transport systems
- Medical imaging without surgical intrusion

9. DISADVANTAGES / RISKS OF EM FIELD EXPOSURE



Strong EM fields can interfere with pacemakers or sensitive electronics

Prolonged exposure to high-intensity EM radiation (e.g., X-rays) can be harmful

Electromagnetic interference (EMI) in critical systems (e.g., aircraft)

Potential health risks from long-term exposure to high-frequency fields (e.g., phone towers)

10. FACTORS THAT DETERMINE THE MAGNITUDE OF AN INDUCED EMF IN A COIL

Speed of rotation of the coil

Cross-sectional area of the coil

Number of turns in the coil



11. EVALUATION (WAEC/NECO PAST QUESTIONS)

Question 1 (WAEC June 2007 Q4)

Explain briefly the principle of operation of an electric motor and state two factors that determine its direction of rotation.

Question 2 (WAEC June 2015 Q6)

(a) Define electromagnetic induction.

(b) State two applications of electromagnetic induction in everyday life.



Question 3 (NECO 2019 Q8)

Describe how a transformer works and mention two devices that use transformers.

Question 4 (WAEC June 2014 Q9a)

Mention two appliances that use electromagnetic fields and describe their working principles.

12. ASSIGNMENT

Define electromagnetic field and describe its two components.

Explain how electromagnetic fields are utilized in:

- (a) MRI
- (b) Induction heating
- (c) Wireless charging

With the aid of diagrams, explain the principle of operation of either a generator or an electric motor.

Discuss two societal concerns related to electromagnetic field exposure and how they are mitigated.

Explain eddy currents

